Multiphase Reaction Engineering (MuRE -- CH3065)
Design Project
Deadline for handing in: 13-04-2026, 09:00AM

Biomass can be used to produce bio-methane by anaerobic digestion. This methane can then be used for steam reforming to produce syngas, a mixture of H_2 and CO. Finally the Fischer-Tropsch process can then be employed to convert syngas into carbon chains of various lengths, which can be used as fuel. The CO_2 emitted upon burning these is taken up by the biomass again, closing the entire loop. Converting syngas to carbon chains is achieved via a complex network of reactions, but can be summarized as follows:

$$(2n+1)H_2 + nCO \xrightarrow{yields} C_nH_{2n+2} + nH_2O \; ; n = 1,2,3 \ldots$$

$$\Delta H = -170 kJmol_{CO}^{-1}$$

In this final assignment, you are asked to make a quantitative design of a Fischer-Tropsch reactor for a feed stream containing 3000 ton CO / day. Also, since the considered reactions are highly exothermic, specific attention to heat removal from the reactor and the temperature homogeneity in the reactor is required.

Use the design approach as given by Krishna and Sie, Chemical Engineering Science 49 (1994) 4029 to find the basic design parameters of your reactor. Show and discuss the effect of the variation of the main design parameters, and determine the optimum layout for your reactor type.

Use the kinetics as given in appendix A. Discuss if you think this is reasonable for your specific reactor and the choices you made for the design parameters.

As always, state which simplifications / assumptions you make and discuss / proof the validity. You should also clearly state the other steps you make in the derivation of your model. Do not simply copy assumptions / models from literature.

The presentations take place on April 13, in the morning. You will have to deliver the report at latest at latest at the start of this session via BrightSpace. In total, this final assignment is estimated to take about 50 hours of work per person. The final presentation should take 10 minutes (excluding questions). You will be informed beforehand who will have to present; students that did not present during CH3061 will certainly have to present.

Criteria for grading the final project include: design approach; justification of assumptions; clarity of description; model complexity; creativity; correctness; references; symbols list; lay-out.

Appendix A: Fischer-Tropsch kinetics

The most widely used kinetics for Fisscher-Tropsch is the Yates and Satterfield expression [1], which follows the Langmuir-Hinshelwood model. It gives the reaction rate based on CO per catalyst mass as:

$$r = \frac{F\ a\ p_{CO}\ p_{H_2}}{(1 + b\ p_{CO})^2}$$

In this equation, F is a catalyst activity multiplication factor that accounts for improvements in catalyst activity since publication of the original parameter values in 1991 [2, 3]; 3 is a reasonable value. p_i is the partial pressure of reactant i, a is the reaction rate coefficient (per unit mass of catalyst) and b is the adsorption coefficient of CO (Eq.17.2), as reported by Maretto and Krishna [4], which can be calculated as:

$$a = a_0 exp\left(\frac{E_a}{R_u}\left(\frac{1}{493.15} - \frac{1}{T}\right)\right)$$

$$b = b_0 exp\left(\frac{\Delta_b H}{R_u}\left(\frac{1}{493.15} - \frac{1}{T}\right)\right)$$

In this expression, R_u is the universal gas constant and T is the temperature in K. The other required values are given in Table A.1. Please note that there is a typo in the original publication of Maretto and Krishna (a minus sign is missing in the exponent of a_0); the values in Table A.1 are correct.

Table A.1 Values required to calculate the temperature dependence of the Yates and Satterfield expression for the Fischer-Tropsch reaction.

Symbol	Meaning	Value	Unit
a_0	Reaction Rate Coefficient at T=493.15 K	$8.88522 \cdot 10^{-3}$	mol s^{-1} kg_{cat}^{-1} bar^{-2}
E_a	Activation energy	$3.737 \cdot 10^4$	J mol^{-1}
b_0	Adsorption coefficient at T=493.15 K	2.226	bar^{-1}
$\Delta_b H$	Adsorption enthalpy	$-6.837 \cdot 10^3$	J mol^{-1}

References:

1. Yates, I.C. and C.N. Satterfield, *Intrinsic kinetics of the Fischer-Tropsch synthesis on a cobalt catalyst.* Energy & Fuels, 1991. **5**(1): p. 168-173.
2. Guettel, R. and T. Turek, *Comparison of different reactor types for low temperature Fischer-Tropsch synthesis: A simulation study.* Chemical Engineering Science, 2009. **64**(5): p. 955-964.
3. Vervloet, D., et al., *Fischer-Tropsch reaction-diffusion in a cobalt catalyst particle: Aspects of activity and selectivity for a variable chain growth probability.* Catalysis Science and Technology, 2012. **2**(6): p. 1221-1233.
4. Maretto, C. and R. Krishna, *Modelling of a bubble column slurry reactor for Fischer-Tropsch synthesis.* Catalysis Today, 1999. **52**(2-3): p. 279-289.

Hints for papers for specific reactors

Multitubular fixed bed (Shell): S. T. Sie, M. M. G. Senden and H. M. H. Van Wechem , Conversion of natural gas to transportation fuels via the shell middle distillate synthesis process (SMDS), Catalysis Today 8, 1991, Pages 371-394

Circulating fluidized bed (Sasol) &Turbulent fluidized bed (Sasol): A. P. Steynberg, R. L. Espinoza, B. Jager and A. C. Vosloo , High temperature Fischer–Tropsch synthesis in commercial practice, Applied Catalysis A: General, 186,1999, Pages 41-54.

Slurry bubble column (Sasol): R. L. Espinoza, A. P. Steynberg, B. Jager and A. C. Vosloo, Low temperature Fischer–Tropsch synthesis from a Sasol perspective, Applied Catalysis A: General, Volume 186, 1999, Pages 13-26.

Airlift loop (TU Delft): Ronald M. de Deugd, Freek Kapteijn, Jacob A. Moulijn, Trends in Fischer–Tropsch Reactor Technology—Opportunities for Structured Reactors, Topics in Catalysis, Volume 26, Issue 1 - 4, 2003, Pages 29 - 39; Sun, S., Liu, C., Wei, W., Bao, X., Hydrodynamics of an annulus airlift reactor, Powder Technology 162 (3), 2006, Pages 201-207.

Staged slurry bubble column (UvA): C. Maretto and R. Krishna, Design and optimisation of a multi-stage bubble column slurry reactor for Fischer–Tropsch synthesis, Catalysis Today, 66, 2001, Pages 241-248.

Slurry bubble column with needle sparger (TU Delft): Hooshyar, N., D. Vervloet, F. Kapteijn, P. J. Hamersma, R. F. Mudde and J. R. van Ommen (2012). "Intensifying the Fischer-Tropsch Synthesis by reactor structuring - A model study." Chemical Engineering Journal 207-208, 2012, Pages 865-870.

Metallic monolith (Milano): Visconti, C.G., Tronconi, E., Groppi, G., Lietti, L., Iovane, M., Rossini, S., Zennaro, R., Monolithic catalysts with high thermal conductivity for the Fischer-Tropsch synthesis in tubular reactors, Chemical Engineering Journal 171 (3), 2011, Pages 1294-1307.

Closed cross flow structure (open channels) (TU Delft): Pangarkar, K., Schildhauer, T.J., Van Ommen, J.R., Nijenhuis, J., Kapteijn, F., Moulijn, J.A., Structured packings for multiphase catalytic reactors, Industrial and Engineering Chemistry Research 47 (10), 2008, Pages 3720-3751.

Closed cross flow structure (packed channels) (TU Delft): Pangarkar, K., Schildhauer, T.J., Van Ommen, J.R., Nijenhuis, J., Kapteijn, F., Moulijn, J.A., Structured packings for multiphase catalytic reactors, Industrial and Engineering Chemistry Research 47 (10), 2008, Pages 3720-3751; Vervloet, D., Kapteijn, F., Nijenhuis, J., Van Ommen, J.R, Process intensification of tubular reactors: Considerations on catalyst hold-up of structured packings, Catalysis Today 216, 2013, Pages 111-116.

Microchannel reactor (NTNU): Holmen, A., Venvik, H.J., Myrstad, R., Zhu, J., Chen, D., Monolithic, microchannel and carbon nanofibers/carbon felt reactors for syngas conversion by Fischer-Tropsch synthesis, Catalysis Today 216, 2013, Pages 150-157.

Structured platelet millireactor (Strasbourg): Liu, Y., Edouard, D., Nguyen, L. D., Begin, D., Nguyen, P., Pham, C., & Pham-Huu, C. (2013). High performance structured platelet milli-

reactor filled with supported cobalt open cell SiC foam catalyst for the Fischer–Tropsch synthesis. Chemical Engineering Journal 222, 2013, pages 265-273.

3D printed open cellular structure (Milano): Visconti, C. G., Panzeri, M., Groppi, G., & Tronconi, E. (2024). Heat transfer intensification in compact tubular reactors with cellular internals: A pilot-scale assessment of highly conductive packed-POCS with skin applied to the Fischer-Tropsch synthesis. Chemical Engineering Journal, 481, 148469.